

Quantum Parallelism & Entanglement

Consider the n -qubit state

$$|0\rangle^{\otimes n} = |00 \dots 0\rangle$$

$$= |0\rangle \otimes |0\rangle \otimes |0\rangle \dots \otimes |0\rangle$$

Now apply $H^{\otimes n}$

$$H^{\otimes n} |0\rangle^{\otimes n} = H(|0\rangle) \otimes H(|0\rangle) \otimes \dots \otimes H(|0\rangle)$$

$$= \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \dots \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$= \frac{1}{\sqrt{2^n}} (|00 \dots 0\rangle + |00 \dots 01\rangle + \dots + |11 \dots 1\rangle)$$

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle$$

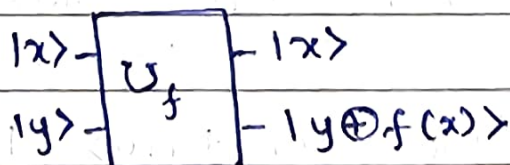
Oracle or Black-box computation

Oracle is a kind of subroutine call.

It is basically a black-box computation of function. Given input x & y , the oracle function U_f is computed as follows.

$$U_f : |x\rangle |y\rangle \longrightarrow |x\rangle |y \oplus f(x)\rangle$$

This can be depicted as



Black-box

If we set $y=0$, output is $f(x)$.

If we set $y=1$, output is complement of $f(x)$.

For example, CNOT is an oracle function. with control bit $f(x)$ and target bit y is flipped if and only if $f(x)=1$. Otherwise no change.

Deutsch Algorithm

Let $f: \{0,1\} \rightarrow \{0,1\}$ be a binary function. The function is either constant $f(0)=f(1)=0$ or $f(0)=f(1)=1$. or balanced $f(0)=0, f(1)=1$ or $f(0)=1, f(1)=0$.

Classical computation requires two function calls, whereas quantum computation requires one computation.

Consider the state

$$|\psi_0\rangle = |0\rangle \otimes |1\rangle$$

Apply $H^{\otimes 2}$

$$|\psi_1\rangle = H^{\otimes 2} |\psi_0\rangle = H|0\rangle \otimes H|1\rangle$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

$$|\psi_1\rangle = \frac{1}{2} \left(\overset{x}{\uparrow} \overset{y}{\uparrow} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \right)$$

Now apply U_f : $|x\rangle|y\rangle \rightarrow |x\rangle|y \oplus f(x)\rangle$

$$\begin{aligned}
 |\psi_2\rangle &= U_f |\psi_1\rangle \\
 &= \frac{1}{2} (|0, f(0)\rangle - |0, \overline{f(0)}\rangle + |1, f(1)\rangle - |1, \overline{f(1)}\rangle)
 \end{aligned}$$

$\xrightarrow{0 \oplus f(0) = f(0)}$ $\xrightarrow{0 \oplus f(1) = f(1)}$
 $\xrightarrow{f(0) \text{ complement}}$ $\xrightarrow{f(1) \text{ complement}}$

$$|\psi_2\rangle = \frac{1}{2} (|0, f(0)\rangle - |0, \overline{f(0)}\rangle + |1, f(1)\rangle - |1, \overline{f(1)}\rangle)$$

$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 H H H H

Apply H on the 1st qubit.

$$\begin{aligned}
 |\psi_3\rangle &= \frac{1}{2\sqrt{2}} \left(\underbrace{(|0\rangle + |1\rangle)}_{(1)} |f(0)\rangle - \underbrace{(|0\rangle + |1\rangle)}_{(1)} |f(\overline{0})\rangle \right. \\
 &\quad \left. + \underbrace{(|0\rangle - |1\rangle)}_{(2)} |f(1)\rangle - \underbrace{(|0\rangle - |1\rangle)}_{(2)} |f(\overline{1})\rangle \right)
 \end{aligned}$$

$$|\psi_3\rangle = \frac{1}{2\sqrt{2}} \left[(|0\rangle + |1\rangle) (|f(0)\rangle - |f(\overline{0})\rangle) + (|0\rangle - |1\rangle) (|f(1)\rangle - |f(\overline{1})\rangle) \right]$$

If function is constant then $f(0) = f(1)$.

$$|\psi_3\rangle = \frac{1}{2\sqrt{2}} |0\rangle (|f(0)\rangle - |f(\overline{0})\rangle)$$

If function is balanced then $f(0) = 0, f(1) = 1$
 i.e., $|f(\overline{0})\rangle = |f(1)\rangle$

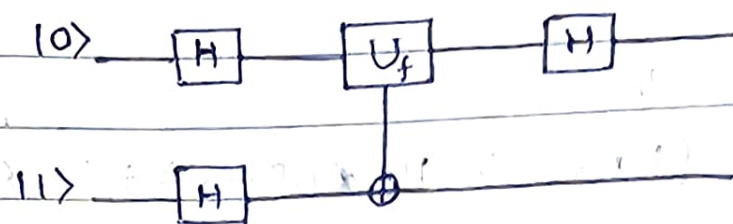
$$|\psi_3\rangle = \frac{1}{\sqrt{2}} |1\rangle (|f(0)\rangle - |f(\overline{0})\rangle)$$

Hence, function is constant if the measurement of 1st qubit is 0. Otherwise it is balanced.

For 1 qubit $\rightarrow$ 2 function calls

For n qubits $\rightarrow (2^{n-1} + 1)$ function calls

Quantum circuit for Deutsch Algorithm.



For 1 qubit, we use $|0\rangle^{\otimes 1} \otimes |1\rangle = |0\rangle \otimes |1\rangle$

2 qubit $|0\rangle^{\otimes 2} \otimes |1\rangle$

n qubit $|0\rangle^{\otimes n} \otimes |1\rangle$

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$H|x\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^x |1\rangle)$$

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{y=0}^1 (-1)^{x \cdot y} |y\rangle$$

For n -qubit

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle$$

Deutsch-Jozsa Algorithm

Consider the binary function

$$f: \{0,1\}^n \rightarrow \{0,1\}$$

such that function is either constant (0 on all inputs or 1 on all inputs) or balanced (returns 1) for half of the input values and (returns 0) for other half.

Step 1: The algorithm begins with $(n+1)$ qubit states.

$$|\psi_0\rangle = |0\rangle^{\otimes n} |1\rangle$$

That is, the 1^{st} n -qubit are each in the state $|0\rangle$ and the final one is $|1\rangle$

Step 2: Apply H to each qubit

$$|\psi_1\rangle = H^{\otimes n} |0\rangle^{\otimes n} H |1\rangle$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Step 3: Apply the oracle, the oracle maps the state $|x\rangle |y\rangle$ to $|x\rangle |y \oplus f(x)\rangle$.

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \frac{1 f(x) - 1 (1 \oplus f(x))}{\sqrt{2}}$$

Step 4: For each x , $f(x)$ is either 0 or 1, then when $f(x) = 0$,

$$|x\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

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$$|x\rangle \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

when $f(x) = 1$

$$|x\rangle = \frac{|1\rangle - |0\rangle}{\sqrt{2}}$$

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Step 4

$$|\psi_3\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle$$

each of n qubits

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(n+1) qubits

Step 5: Apply H to 1st n-qubits

$$|\psi_4\rangle = \frac{1}{2^n} \sum_{x=0}^{2^n-1} \sum_{y=0}^{2^n-1} (-1)^{f(x)} (-1)^{x \cdot y} |y\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
 $H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

$$= \left(\frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |0\rangle^{\otimes n} \right) + \frac{1}{2^n} \sum_{y=1}^{2^n-1} (-1)^{f(x)} H|x\rangle = \frac{1}{\sqrt{2}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle$$

$|x|^2$ $(-1)^{x \cdot y} |y\rangle$ $\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$ For n qubits,
 $H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{y=0}^{2^n-1} (-1)^{x \cdot y} |y\rangle$

(n+1) qubits

Probability of

getting 1st n-qubits as zero, $P(|0\rangle^{\otimes n}) = \left| \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \right|^2$

Now we have 2 cases when $f(x)$ is constant

Case I: $f(x) = 0 \quad \forall x \in \{0, \dots, 2^n-1\}$

$$(-1)^{f(x)} = 1 \quad \forall x$$

$$\left| \frac{1}{2^n} \sum_{x=0}^{2^n-1} 1 \right|^2 = \frac{1}{2^n} \cdot 2^n = 1$$

$$\sum_{i=l}^u = u - l + 1$$

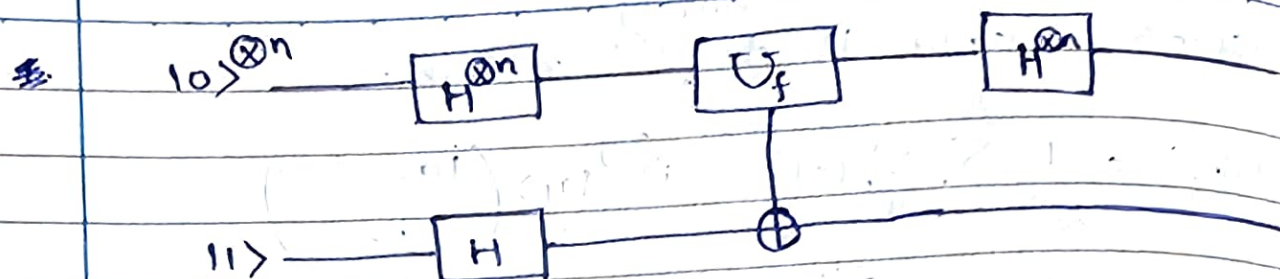
Case II: $f(x) = 1 \quad \forall x$

$$(-1)^{f(x)} = -1 \quad \forall x$$

$$\left| \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1) \right|^2 = \left| \frac{1}{2^n} \cdot (-2^n) \right|^2 = |-1|^2 = 1$$

Otherwise $f(x)$ is balanced.

i.e., $P(|0\rangle^{\otimes n}) \neq 1$



→ Quantum circuit for Deutsch-Jozsa Algorithm.

1. Take $n=2$ Deutsch-Jozsa Algorithm. Find final state $|\psi_4\rangle$ & evaluate the measurement outcomes & their probabilities for each case.

i) $f(00) = f(01) = f(10) = f(11) = 1$ (constant) → Ans: $P(00) = 1$

(HW) ii) $f(00) = 0, f(01) = f(10) = 1, f(11) = 0$ (balanced)

→ Ans: $P(00) = 0$

Soln i) Step 1: $|\psi_0\rangle = |0\rangle^{\otimes 2} |1\rangle$
 $= |00\rangle |1\rangle$

Step 2: Apply H to each 3 qubits.

$$|\psi_1\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$H^{\otimes 2} |00\rangle = H|0\rangle \otimes H|0\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

Step 3: $|\psi_3\rangle = |\psi_2\rangle = \frac{1}{2} (-|00\rangle - |01\rangle - |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$

$|x\rangle \rightarrow (-1)^{f(x)} |x\rangle$

$f(x) = 1$ for all x in constant

first 2 qubits

3rd qubit

Step 4: Apply H to each 1st 2-qubits.

$$= \frac{1}{2} \left(\frac{-1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) - \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) - \frac{1}{2} (|00\rangle + |01\rangle - |10\rangle - |11\rangle) - \frac{1}{2} (|00\rangle - |01\rangle - |10\rangle + |11\rangle) \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$= \frac{1}{4} \left(-|00\rangle - |01\rangle - |10\rangle - |11\rangle - |00\rangle + |01\rangle - |10\rangle + |11\rangle - |00\rangle - |01\rangle + |10\rangle + |11\rangle - |00\rangle + |01\rangle + |10\rangle - |11\rangle \right) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$= \frac{1}{4} (-4|00\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$= \underbrace{-100}_{\alpha} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$\rightarrow P(00) = |-1|^2 = 1.$$

(Probability of getting 1st 2-qubits 0)

ii) $f(00)=0$, $f(01)=f(10)=1$, $f(11)=0$ (balanced)

Step 1: $|\psi_0\rangle = |0\rangle \otimes^2 |1\rangle$

$$|\psi_0\rangle = |00\rangle |1\rangle$$

Step 2: Apply H to each 3 qubits

$$|\psi_1\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Step 3: Apply Oracle

$$|\psi_2\rangle = |\psi_3\rangle = \frac{1}{2} \underbrace{(|00\rangle - |01\rangle - |10\rangle + |11\rangle)}_{\substack{\uparrow \uparrow \\ H \ H}} \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Step 4: Apply H to each first 2 qubits.

$$|\psi_4\rangle = \frac{1}{4} (|00\rangle + |01\rangle + |10\rangle + |11\rangle - |00\rangle + |01\rangle - |10\rangle + |11\rangle - |00\rangle - |01\rangle + |10\rangle + |11\rangle + |00\rangle - |01\rangle - |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$|\psi_4\rangle = \frac{4}{4} |11\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

The probability of getting 1st 2 qubits zero is

$$\underline{P(00) = 0} \quad (\text{balanced})$$

$$\text{iii) } f(00) = 0, f(01) = f(10) = f(11) = 1$$

Ans: neither constant nor balanced.

$$|\Psi_4\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$P(00) = \frac{1}{2}$$

Berstein - Vazirani Algorithm

Consider the binary function.

$$f: \{0,1\}^n \rightarrow \{0,1\}$$

such that

$$f(x) = s \cdot x \pmod{2}$$

where x is an n -bit input string. $(x_1, x_2, \dots, x_n)$

s is an unknown n -bit secret string.

$$(s_1, s_2, \dots, s_n)$$

$$s \cdot x = s_1 x_1 \oplus s_2 x_2 \oplus \dots \oplus s_n x_n$$

Determine secret string s .

1. Consider the Bernstein - Vazirani algorithm with $n=2$ & $s=01$. Show that the measurement outcome of the 1st 2 qubits is $s=01$.

$$f(x) = s \cdot x = s_1 x_1 \oplus s_2 x_2 \quad (s=01)$$

$$f(00) = 0 \cdot x_1 \oplus 1 \cdot x_2 = 0 \cdot 0 \oplus 1 \cdot 0 = 0$$

$$f(01) = 0 \cdot x_1 \oplus 1 \cdot x_2 = 1$$

$$f(10) = 0 \cdot x_1 \oplus 0 \cdot x_2 = 0$$

$$f(11) = 0 \cdot x_1 \oplus 1 \cdot x_2 = 1$$

Step 1: $|\psi_0\rangle = |0\rangle^{\otimes 2} |1\rangle$

$$|\psi_0\rangle = |00\rangle |1\rangle$$

Step 2: $|\psi_1\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$

Step 3: Apply oracle

$$|\psi_2\rangle = |\psi_3\rangle = \frac{1}{2} (|00\rangle - |01\rangle + |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

Step 4: Apply H to 1st 2 qubits.

$$|\psi_4\rangle = \frac{1}{4} (|00\rangle + |01\rangle + |10\rangle + |11\rangle - |00\rangle + |01\rangle - |10\rangle + |11\rangle + |00\rangle + |01\rangle - |10\rangle - |11\rangle - |00\rangle + |01\rangle + |10\rangle - |11\rangle) \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$|\psi_4\rangle = |01\rangle \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

$$\underline{s = 01}$$